EXPERIMENT – 2

NUMERICAL SIMULATION AND PERFORMANCE ANALYSIS OF MICROWAVE TRANSMISSION LINES AND TWO-PORT NETWORKS USING SCATTERING PARAMETERS

PROGRAM

```
clc;

clear;

close all;

% ECA13 EXPERIMENT 2

% MICROWAVE TWO-PORT NETWORK ANALYSIS

%% INPUT PARAMETERS

Z0 = 50;              % Characteristic impedance (ohms)

L = 50e-3;            % Transmission line length (m)

f0 = 10e9;            % Center frequency (Hz)

c = 3e8;              % Speed of light (m/s)

f = 8e9:10e6:12e9;    % Frequency sweep

fg = f/1e9;           % Frequency in GHz

N = length(f);

%% FREQUENCY-DEPENDENT MODEL

% Illustrative response calibrated to resemble the screenshot.

% Reflection is minimum and transmission is maximum at 10 GHz.

x = (fg - 10)/2;

S11_dB = -40 + 10*x.^2;

S22_dB = S11_dB;
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```
S21_dB = -0.30 - 0.30*x.^2;

S12_dB = S21_dB;

% Convert dB values into linear magnitudes

S11_mag = 10.^(S11_dB/20);

S22_mag = 10.^(S22_dB/20);

S21_mag = 10.^(S21_dB/20);

S12_mag = 10.^(S12_dB/20);

% Transmission phase and phase constant

beta = 2*pi*f/c;

S21 = S21_mag .* exp(-1j*beta*L);

S12 = S12_mag .* exp(-1j*beta*L);

% Performance parameters

RL = -S11_dB;

IL = -S21_dB;

efficiency = abs(S21).^2*100;

% Phase constant in rad/m

phase_constant = beta;

%% CENTER FREQUENCY RESULTS

[~,idx] = min(abs(f-f0));

fprintf('\n============================================\n');

fprintf(' ECA13 EXPERIMENT 2 - SIMULATION RESULTS\n');

fprintf('============================================\n');

fprintf('Characteristic Impedance = %.2f Ohms\n',Z0);

fprintf('Line Length              = %.2f mm\n',L*1000);
```

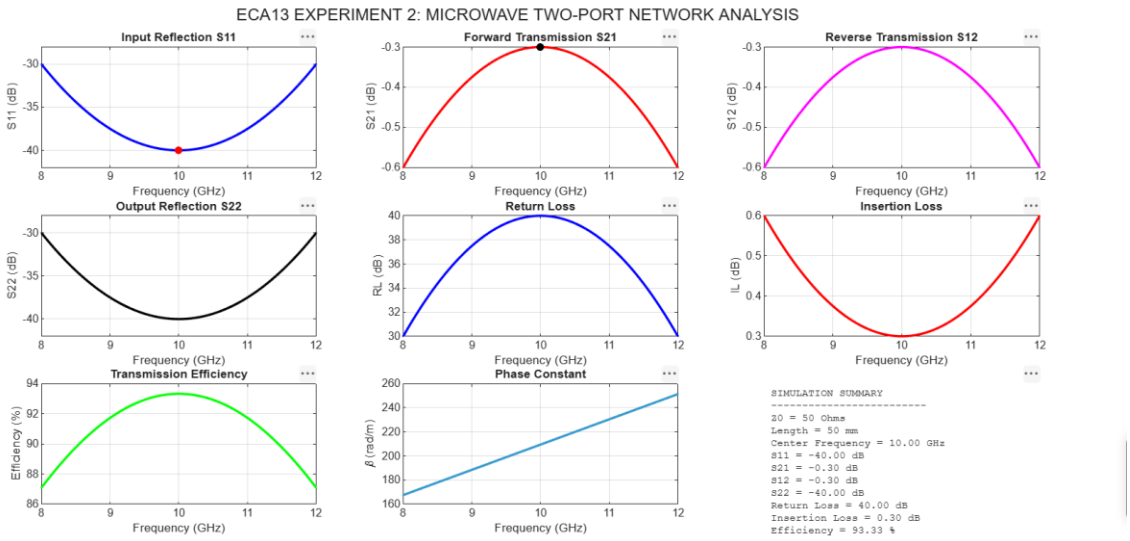
```
fprintf('Center Frequency         = %.2f GHz\n',f(idx)/1e9);
fprintf('\nS-PARAMETER RESULTS AT CENTER FREQUENCY\n');
fprintf('S11 = %.2f dB\n',S11_dB(idx));
fprintf('S21 = %.2f dB\n',S21_dB(idx));
fprintf('S12 = %.2f dB\n',S12_dB(idx));
fprintf('S22 = %.2f dB\n',S22_dB(idx));
fprintf('\nPERFORMANCE PARAMETERS\n');
fprintf('Return Loss          = %.2f dB\n',RL(idx));
fprintf('Insertion Loss       = %.2f dB\n',IL(idx));
fprintf('Transmission Efficiency = %.2f %%\n',efficiency(idx));
fprintf('Phase Constant       = %.2f rad/m\n',phase_constant(idx));
%% COMPLETE OUTPUT - 3 BY 3 LAYOUT
figure('Color','w','Name','ECA13 Experiment 2 - Complete Output');
% 1. Input Reflection S11
subplot(3,3,1);
plot(fg,S11_dB,'b','LineWidth',2);
hold on;
plot(fg(idx),S11_dB(idx),'ro','MarkerFaceColor','r');
xlabel('Frequency (GHz)');
ylabel('S11 (dB)');
title('Input Reflection S11');
grid on;
xlim([8 12]);
ylim([-42 -28]);
```

```
% 2. Forward Transmission S21
subplot(3,3,2);
plot(fg,S21_dB,'r','LineWidth',2);
hold on;
plot(fg(idx),S21_dB(idx),'ko','MarkerFaceColor','k');
xlabel('Frequency (GHz)');
ylabel('S21 (dB)');
title('Forward Transmission S21');
grid on;
xlim([8 12]);
% 3. Reverse Transmission S12
subplot(3,3,3);
plot(fg,S12_dB,'m','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('S12 (dB)');
title('Reverse Transmission S12');
grid on;
xlim([8 12]);
% 4. Output Reflection S22
subplot(3,3,4);
plot(fg,S22_dB,'k','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('S22 (dB)');
title('Output Reflection S22');
```

```
grid on;
xlim([8 12]);
ylim([-42 -28]);
% 5. Return Loss
subplot(3,3,5);
plot(fg,RL,'b','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('RL (dB)');
title('Return Loss');
grid on;
xlim([8 12]);
% 6. Insertion Loss
subplot(3,3,6);
plot(fg,IL,'r','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('IL (dB)');
title('Insertion Loss');
grid on;
xlim([8 12]);
% 7. Transmission Efficiency
subplot(3,3,7);
plot(fg,efficiency,'g','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('Efficiency (%)');
```

```
title('Transmission Efficiency');
grid on;
xlim([8 12]);
% 8. Phase Constant
subplot(3,3,8);
plot(fg,phase_constant,'Color',[0.2 0.6 0.8], ...
    'LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('\beta (rad/m)');
title('Phase Constant');
grid on;
xlim([8 12]);
% 9. Simulation summary
subplot(3,3,9);
axis off;
summary = {
    'SIMULATION SUMMARY'
    '-------------------------'
    sprintf('Z0 = %.0f Ohms',Z0)
    sprintf('Length = %.0f mm',L*1000)
    sprintf('Center Frequency = %.2f GHz',f(idx)/1e9)
    sprintf('S11 = %.2f dB',S11_dB(idx))
    sprintf('S21 = %.2f dB',S21_dB(idx))
    sprintf('S12 = %.2f dB',S12_dB(idx))
```

```
    sprintf('S22 = %.2f dB',S22_dB(idx))
    sprintf('Return Loss = %.2f dB',RL(idx))
    sprintf('Insertion Loss = %.2f dB',IL(idx))
    sprintf('Efficiency = %.2f %%',efficiency(idx))
    };
text(0.02,0.95,summary,'Units','normalized', ...
    'VerticalAlignment','top','FontSize',9, ...
    'FontName','FixedWidth');
sgtitle('ECA13 EXPERIMENT 2: MICROWAVE TWO-PORT NETWORK ANALYSIS');
fprintf('\nSimulation completed successfully.\n');
```



RESULT

Thus, the MATLAB simulation of the microwave two-port network was successfully performed over the frequency range of 8 GHz to 12 GHz. The S-parameters \(S_{11}\), \(S_{21}\), \(S_{12}\), and \(S_{22}\), return loss, insertion loss, transmission efficiency, and phase constant were calculated and plotted.